

MDS-003 — Material System

External publication draft

MDS-003 — Material System

Materials are not textures. They are physical behaviours through which the user's entertainment becomes part of the interface.

Purpose

The Mosaic Design Language established:

- Why Mosaic exists.
- How it thinks.
- How it behaves.
- How understanding is organised.

The Design System has established:

- Design Tokens
- Colour Architecture

MDS-003 defines how those concepts become **physical interface materials**.

Unlike traditional design systems, Mosaic does not treat materials as decorative surfaces.

Materials are considered physical participants within the user's entertainment World.

They receive:

- Runtime Atmosphere
- Refraction
- Light
- Hierarchy
- Interaction

and transform those inputs into a coherent visual experience.

Relationship to Previous Specifications

[text]

Vision

↓

Principles

↓

Mental Model

↓

Interaction

↓

Composition

↓

Tokens

↓

Colour

↓

Material

↓

Typography

↓

Components

The Material System consumes:

- Design Tokens
- Semantic Colours
- Runtime Atmosphere

It produces:

- Physical Surfaces
- Acrylic Behaviour
- Refraction
- Depth
- Environmental Response

Scope

This specification defines:

- Material Philosophy
- Material Hierarchy
- Canvas
- Acrylic
- Hero Material
- Overlay Material
- Refraction
- UV Mapping
- Light Transport
- Material Resolution
- Runtime Material Behaviour

This specification intentionally does **not** define:

- Typography
- Components
- Motion Curves
- Layout
- Interaction

Those systems consume Materials.

They do not define them.

Core Question

MDS-003 exists to answer one question.

How should the interface physically exist?

Not:

How should it be coloured?

Material Statement

Within Mosaic:

Materials communicate physical presence rather than visual decoration.

Every material should answer:

- What is this?
- How does light interact with it?
- How does it relate to the user's entertainment?

before it answers:

- How does it look?

Material Responsibilities

The Material System separates responsibilities into independent layers.

[text]

Material Identity

↓

Light Behaviour

↓

Runtime Atmosphere

↓

Refraction

↓

Presentation

Each layer contributes one responsibility.

No layer duplicates another.

Expected Outcome

After reading MDS-003 contributors should understand:

- why Mosaic uses Acrylic
- how Runtime Atmosphere interacts with materials
- how Refraction works
- how UV mapping supports adaptation
- how surfaces establish hierarchy
- how future rendering systems should implement these ideas

without discussing specific rendering technologies.

Repository Structure

[text]
design/

■■■ mds/

■■■ MDS-003 Material System/

README.md

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references.md

Dependencies

Required reading:

- MDL-001 → MDL-005
- MDS-001 Design Token Architecture
- MDS-002 Colour System

Downstream specifications:

- MDS-004 Typography
- MDS-005 Motion System
- MDS-006 Composition Engine
- MDS-007 Tile Framework
- MDS-008 Component Library

Review Status

Status

Draft

Owner

Lead Design Systems Architect

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00-document-control.md

Document Control

Document Information

Property	Value
Document ID	MDS-003
Title	Mosaic Design System — Material System
Classification	Internal
Status	Draft
Version	0.1
Owner	Lead Design Systems Architect
Parent Specifications	MDL-001 → MDL-005, MDS-001, MDS-002
Repository	/design/mds/MDS-003 Material System/

Purpose

MDS-003 defines the Material System used throughout Mosaic.

The Material System transforms abstract design concepts into physical surfaces that users perceive as existing within the same environment as their entertainment.

Unlike conventional interface frameworks where materials are decorative styling, Mosaic materials are considered part of the platform's behavioural language.

Materials communicate:

- hierarchy
- physicality
- environmental lighting
- depth
- atmosphere
- continuity

The Material System therefore becomes one of the defining characteristics of the Mosaic experience.

Authority

MDS-003 governs:

- Material hierarchy
- Surface behaviour
- Acrylic

- Canvas
- Hero materials
- Overlay materials
- Runtime material adaptation
- Refraction
- Light transport
- Material resolution

This specification intentionally does **not** govern:

- Typography
- Components
- Motion timing
- Layout
- Interaction behaviour

Those specifications consume materials.

They do not define them.

Relationship To MDS

The Material System extends the Design Token Architecture and Colour System.

[mermaid]
flowchart TD

```
Vision
Vision --> Principles
Principles --> MentalModel
MentalModel --> Interaction
Interaction --> Composition
Composition --> Tokens
Tokens --> Colour
Colour --> Materials
Materials --> Components
Components --> Presentation
```

The Material System consumes:

- Semantic Tokens
- Runtime Atmosphere
- Composition

It produces:

- physical surfaces
- environmental depth
- acrylic behaviour
- refraction

Design Intent

Most design systems describe surfaces.

Mosaic intentionally describes materials.

A surface answers:

"What colour is this?"

A material answers:

"How does this exist?"

Materials should behave as though they occupy physical space.

They should react to:

- light
- atmosphere
- movement
- surrounding content

The result should feel less like software and more like an environment.

Reader Expectations

Before reading this specification contributors should already understand:

- MDL-001 Vision
- MDL-002 Principles
- MDL-003 Mental Model
- MDL-004 Interaction Model
- MDL-005 Composition Model
- MDS-001 Design Token Architecture
- MDS-002 Colour System

MDS-003 assumes those concepts already exist.

Its responsibility is to define their physical expression.

Architectural Scope

The Material System defines:

- Material Identity
- Surface Hierarchy

- Acrylic Behaviour
- Refraction
- Light Response
- Runtime Adaptation

It intentionally avoids implementation technologies such as:

- CSS
- Flutter
- Metal
- Vulkan
- WebGL
- Canvas
- Skia

These are implementation concerns.

The Material System defines only the architectural model.

Stability

Expected lifetime.

Artefact	Expected Lifetime
Shader Implementation	Months
Rendering Backend	Months
Material Algorithms	Years
Material Hierarchy	Years
Material Philosophy	Decades

Rendering technologies are expected to evolve.

Material behaviour should remain recognisably Mosaic.

Success Criteria

MDS-003 succeeds when:

- materials feel physically believable
- runtime atmosphere subtly influences surfaces
- refraction strengthens immersion without distraction
- hierarchy remains immediately understandable
- artwork appears to illuminate the interface

- components feel like objects within one coherent environment

Users should not consciously admire the materials.

They should simply feel that the interface belongs naturally beside the entertainment it presents.

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Dependencies

- MDL-001 → MDL-005
- MDS-001
- MDS-002

Supersedes

None.

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01-material-philosophy.md

Material Philosophy

Purpose

Before defining Acrylic, Refraction or Material Hierarchy, contributors must first understand what a **Material** represents within Mosaic.

Many interface frameworks treat materials as visual effects.

Glass.

Blur.

Opacity.

Shadow.

These are rendering techniques.

They are **not** materials.

Within Mosaic, a Material is a behavioural concept.

It defines how part of the interface physically exists inside the user's entertainment World.

Philosophy Statement

Materials should behave as though they are physical objects illuminated by the user's entertainment rather than software panels layered on top of it.

Everything within the Material System derives from this statement.

Why Materials Exist

Entertainment already possesses:

- colour
- lighting
- atmosphere
- texture
- emotion

Traditional interfaces frequently cover these qualities with opaque panels.

Mosaic intentionally avoids this.

Instead, the interface should feel like it occupies the same environment as the content.

Not a different one.

The Material System exists to make software feel less like software.

Material Is Behaviour

One of the defining architectural ideas of Mosaic is:

Materials are behaviours.

Not appearances.

Acrylic is not simply:

- blur
- transparency
- opacity

Acrylic behaves like a material.

It:

- receives light
- diffuses light
- refracts colour
- preserves hierarchy
- supports interaction

Rendering merely communicates these behaviours.

Materials Exist Between

Materials intentionally exist between:

Entertainment

↓

Atmosphere

↓

Interface

They form the physical medium through which entertainment influences the interface.

Without materials:

Artwork

↓

Interface

With materials:

Artwork

↓

Atmosphere

↓

Material

↓

Interface

This additional layer creates a far more natural experience.

Materials Are Environmental

Materials belong to the environment.

Not to components.

Poor.

Button

↓

Blur

Preferred.

Environment

↓

Material

↓

Button Exists Inside Material

Components inherit the behaviour of their surrounding material.

They should not define materials independently.

Materials Respond To Light

The defining property of every Mosaic material is:

Light interaction.

Materials should respond to:

- Runtime Atmosphere
- Hero artwork
- environmental luminance
- surrounding hierarchy

They should not simply display static colours.

Static materials quickly become lifeless.

Materials Preserve Hierarchy

Materials should reinforce hierarchy.

Not replace it.

Example.

Canvas

↓

Acrylic

↓

Hero

Each material communicates increasing:

- importance
- physical presence
- environmental influence

without requiring additional decoration.

Materials Are Restrained

One of the greatest dangers of advanced material systems is excess.

Too much blur.

Too much glow.

Too much translucency.

Too much reflection.

Mosaic intentionally adopts restraint.

Materials should be noticed emotionally.

Not intellectually.

Users should rarely think:

"That acrylic looks impressive."

Instead they should feel:

"This interface belongs here."

Materials Support Entertainment

Materials exist to support entertainment.

Not compete with it.

Artwork should remain:

- sharper
- richer
- emotionally dominant

Materials should quietly reinforce:

- atmosphere
- hierarchy
- continuity

The interface should recede naturally behind content.

Materials And Runtime Atmosphere

Runtime Atmosphere provides environmental light.

Materials determine how that light behaves.

Example.

Artwork

↓

Runtime Atmosphere

↓

Acrylic

↓

Refraction

↓

Presentation

Atmosphere supplies energy.

Materials determine physical response.

This separation allows both systems to evolve independently.

Materials And Movement

Movement should affect materials naturally.

Examples.

Moving Hero.

↓

Refraction subtly shifts.

Changing Focus.

↓

Atmosphere gradually redistributes.

Changing Domain.



Environmental lighting evolves.

Materials should reinforce behavioural continuity.

Not introduce independent visual effects.

Materials Across Themes

Light and Dark themes should alter material behaviour.

Not material identity.

Light Theme.

- softer diffusion
- brighter scattering
- lighter acrylic

Dark Theme.

- deeper diffusion
- richer reflections
- stronger perceived depth

Users should recognise the same materials in every theme.

Only the environmental lighting changes.

Materials Are Device Independent

Material philosophy should remain identical across:

- Web
- Desktop
- Mobile
- Television
- Future clients

Different rendering engines may implement:

- shaders
- filters
- translucency

differently.

The perceived material should remain recognisably Mosaic.

Good Examples

Hero

Artwork softly illuminates surrounding acrylic.

The Hero remains dominant.

The material quietly supports it.

Navigation

Navigation remains largely neutral.

Very subtle environmental influence.

Orientation remains stronger than atmosphere.

Playback

Controls inherit gentle reflected light.

Video remains visually dominant.

Nothing distracts from entertainment.

Anti-patterns

Frosted Glass Everywhere

Every surface becomes heavily blurred.

Hierarchy disappears.

Decorative Glow

Glow exists because it looks impressive.

No additional understanding is created.

Flat Panels

Materials become static coloured rectangles.

Atmosphere disappears.

Heavy Transparency

Background content overwhelms foreground readability.

Accessibility weakens.

Material Philosophy Model

[mermaid]
flowchart TD

```
Entertainment
Entertainment --> RuntimeAtmosphere
RuntimeAtmosphere --> Materials
Materials --> Refraction
Refraction --> Presentation
Presentation
Presentation --> UserPerception
```

Materials are the physical medium through which entertainment influences perception.

Relationship To Future Chapters

The remaining chapters define the individual materials within Mosaic.

These include:

- Canvas
- Acrylic
- Hero Material
- Overlay Material
- Refraction
- UV-indexed Refraction
- Light Transport
- Runtime Material Resolution

Every implementation should reinforce the philosophy established here.

Summary

Materials are not decorative styling.

They are the physical expression of the Mosaic Design Language.

Their responsibility is to create the feeling that:

- the interface occupies the same world as the entertainment,
- light naturally flows through the experience,
- hierarchy remains physically believable,

- and software quietly fades into the background.

When successful, users should remember the atmosphere...

...not the material that created it.

Review Status

Status

Draft

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02-material-hierarchy.md

Material Hierarchy

Purpose

Every physical object communicates importance.

A sheet of paper does not feel the same as polished acrylic.

Glass does not feel the same as carved stone.

Likewise, every material within Mosaic should communicate its role through its physical behaviour.

The Material Hierarchy defines the architectural relationship between every material used throughout the platform.

It answers one question.

"How physically present should this part of the interface feel?"

Definition

Within MDS, **Material Hierarchy** is defined as:

The ordered system through which materials communicate hierarchy using physical behaviour rather than decoration.

Material Hierarchy is independent from:

- colour
- typography
- components
- layout

It is an environmental system.

Why Hierarchy Exists

Without Material Hierarchy:

Every surface behaves identically.

Every object appears equally important.

Users must rely entirely upon colour or layout to understand hierarchy.

Mosaic intentionally distributes hierarchy across multiple systems.

Including:

- Composition
- Colour

- Typography
- Motion
- Materials

Materials reinforce understanding.

They never create it independently.

Material Layers

The Mosaic Material System defines five primary layers.

[text]

Canvas

↓

Surface

↓

Acrylic

↓

Hero

↓

Overlay

Each layer increases perceived physical presence.

Layer 01

Canvas

Purpose.

Represent the environment.

The Canvas should feel:

- calm
- stable
- neutral
- distant

It should receive only subtle influence from Runtime Atmosphere.

The Canvas exists to support everything above it.

It should rarely attract attention itself.

Layer 02

Surface

Purpose.

Group related understanding.

Examples include:

- tiles
- panels
- collections
- shelves

Surfaces should possess:

- gentle separation
- minimal elevation
- restrained material response

Surfaces organise information.

They should not dominate it.

Layer 03

Acrylic

Purpose.

Create physical presence.

Acrylic introduces:

- translucency
- refraction
- environmental lighting
- perceived depth

Acrylic should feel:

- premium
- contemporary
- substantial

Importantly...

Acrylic is **not glass**.

Glass disappears.

Acrylic possesses presence.

It occupies space while remaining connected to its surroundings.

This distinction forms one of the defining visual characteristics of Mosaic.

Layer 04

Hero

Purpose.

Communicate current importance.

Hero Materials receive the strongest influence from:

- Runtime Atmosphere
- Artwork
- Refraction

Hero Materials should feel:

- illuminated
- immersive
- emotionally connected

without becoming visually dominant.

The entertainment remains the emotional centre.

The Hero Material quietly supports it.

Layer 05

Overlay

Purpose.

Temporarily communicate interaction.

Examples include:

- playback controls
- dialogs
- search
- menus

Overlay Materials should prioritise:

- readability
- separation

- focus

Runtime Atmosphere influence should reduce while overlays remain active.

Interaction always has higher priority than atmosphere.

Physical Presence

Every material possesses perceived physical weight.

Conceptually.

[text]
Canvas

↓

Very Light

Surface

↓

Light

Acrylic

↓

Medium

Hero

↓

High

Over lay

↓

Highest

Physical weight influences:

- depth
- refraction
- diffusion
- translucency
- environmental response

It should not be confused with visual brightness.

Material Relationships

Materials should behave like members of the same physical family.

Poor.

Glass

Metal

Paper

Plastic

Each behaves independently.

Preferred.

Canvas

↓

Surface

↓

Acrylic

↓

Hero

↓

Overlay

Every material shares:

- lighting model
- refraction language
- environmental response

The differences are gradual.

Not absolute.

Hierarchy Is Behaviour

Material Hierarchy should evolve alongside Composition.

Example.

Hero Changes

↓

Material Importance Changes

↓

Lighting Changes

↓

Refraction Changes

Materials should never change independently from behavioural hierarchy.

They respond to understanding.

Not interface events.

Hierarchy Across Themes

The hierarchy remains identical across themes.

Light Theme.

Canvas

↓

Paper-like

Dark Theme.

Canvas

↓

Slate-like

Hero remains Hero.

Acrylic remains Acrylic.

Only physical interpretation changes.

Hierarchy Across Devices

Different rendering technologies may communicate Material Hierarchy differently.

Desktop.

↓

High-quality acrylic.

Mobile.

↓

Reduced translucency.

Television.

↓

Greater perceived depth.

The behavioural hierarchy remains unchanged.

Runtime Behaviour

Runtime Atmosphere should influence each layer differently.

Material	Atmosphere Influence
Canvas	Very Low
Surface	Low
Acrylic	Medium

Material	Atmosphere Influence
Hero	High
Overlay	Minimal

This prevents atmosphere from overwhelming the interface.

The user's attention naturally remains on current content.

Plugins

Extensions never define materials.

Plugins contribute:

- Information
- Relationships
- Expressions

The Material System determines:

- physical presence
- lighting
- hierarchy
- atmosphere

Every extension therefore inherits one coherent physical language.

Good Examples

Continue Watching

Canvas remains neutral.

Hero gently illuminated.

Supporting tiles receive subtle acrylic.

Navigation remains restrained.

The hierarchy is immediately understandable.

Playback

Video dominates.

Controls appear using Overlay Material.

Atmosphere reduces.

Readability increases.

Interaction feels effortless.

Administration

Canvas becomes calmer.

Atmosphere becomes minimal.

Materials remain recognisably Mosaic while supporting productivity rather than immersion.

Anti-patterns

Everything Acrylic

Every surface possesses identical material.

Hierarchy disappears.

Decorative Depth

Elevation exists purely because it looks modern.

No additional understanding is communicated.

Theme Materials

Different themes invent different material hierarchies.

Users lose physical consistency.

Plugin Materials

Extensions introducing their own material behaviour.

The physical language fragments.

Material Hierarchy Model

[mermaid]
flowchart TD

```
Canvas
Canvas --> Surface
Surface --> Acrylic
Acrylic --> Hero
Hero --> Overlay
Overlay
Overlay --> Presentation
```

Each material increases physical presence while remaining part of one coherent environmental system.

Relationship To Future Chapters

The following chapters define each material individually.

Including:

- Canvas
- Acrylic
- Hero Material
- Overlay Material
- Refraction
- UV-indexed Refraction
- Light Transport

This chapter establishes only the hierarchy connecting them.

Summary

Material Hierarchy defines how physical presence communicates importance.

It allows users to understand hierarchy before consciously analysing colour or layout.

Every material belongs to one coherent physical family.

The interface should feel constructed from one environmental language rather than several unrelated visual effects.

That physical coherence is one of the defining characteristics of the Mosaic Material System.

Review Status

Status

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03-canvas.md

Canvas

Purpose

The Canvas is the foundation of every Mosaic experience.

Everything else exists upon it.

Unlike ordinary application backgrounds, the Canvas is **not** empty space.

It is an environmental material.

Its responsibility is to create a calm, stable world in which entertainment can exist.

The Canvas should quietly disappear from conscious attention while subtly reinforcing the atmosphere established by the current entertainment.

Definition

Within MDS, **Canvas** is defined as:

The foundational material representing the environmental space in which every Mosaic Composition exists.

Canvas is not:

- wallpaper
- artwork
- decoration
- branding

Canvas represents the environment itself.

Philosophy

A museum wall exists to support paintings.

It should not compete with them.

Likewise, the Mosaic Canvas exists to support entertainment.

It should create:

- calm
- space
- depth
- confidence

without asking for attention.

The Canvas should never become the Hero.

Behaviour

The Canvas should always feel:

- stable
- continuous
- restrained
- physically distant

Unlike Acrylic, it should not appear to float above the content.

Instead it should behave as the physical environment surrounding every interface element.

Canvas Responsibilities

The Canvas has five responsibilities.

1. Establish Environment

The Canvas defines the overall visual environment before any interface elements appear.

2. Support Atmosphere

Runtime Atmosphere subtly influences the Canvas.

The influence should remain extremely restrained.

3. Provide Contrast

Foreground materials must always remain distinguishable from the Canvas.

4. Preserve Calm

The Canvas should reduce visual noise.

Never increase it.

5. Maintain Continuity

The Canvas should remain visually continuous while the rest of the Composition evolves.

Runtime Atmosphere

Canvas participates in Runtime Atmosphere.

However...

It possesses the **lowest atmospheric weighting** of every material.

Conceptually.

Artwork

↓

Atmosphere

↓

Canvas

↓

Very Low Influence

The user should perceive the atmosphere emotionally.

Not consciously notice coloured backgrounds.

Neutrality

The Canvas should begin from a neutral foundation.

Examples.

Light Theme.

Warm Paper

Dark Theme.

Deep Slate

Future implementations may refine these neutrals.

The underlying philosophy remains unchanged.

Neutrality provides room for:

- artwork
- Hero materials
- Acrylic
- typography

to become visually expressive.

Texture

The Canvas may contain an extremely subtle physical texture.

Examples include:

- fine grain
- paper fibre
- soft diffusion
- micro-noise

Texture should exist to prevent the interface feeling digitally flat.

It should never become recognisable as a pattern.

Users should feel texture rather than see it.

Lighting

The Canvas should receive indirect environmental light.

Unlike Hero Materials, which respond visibly to atmosphere...

Canvas should respond almost imperceptibly.

Example.

Sci-Fi Artwork

↓

Very subtle cool environmental shift.

Fantasy Artwork

↓

Very subtle warm environmental shift.

The effect should resemble ambient room lighting.

Not recolouring.

Canvas And Motion

The Canvas should rarely move.

Instead...

Materials move within the Canvas.

The environment remains stable.

This stability significantly improves orientation.

If the Canvas appears to move independently from the user's World, continuity weakens.

Canvas And Composition

Canvas exists beneath every Composition.

Conceptually.

[text]
Canvas

↓

Composition

↓

Materials

↓

Content

The Composition may evolve continuously.

Canvas should remain behaviourally stable.

This stability creates the feeling that the user remains inside one coherent environment.

Canvas Across Themes

Light Theme.

Canvas behaves like softly illuminated paper.

Dark Theme.

Canvas behaves like deep architectural slate.

OLED Theme.

Canvas may approach true black where appropriate.

Despite these differences...

Users should always recognise:

Canvas

The material identity remains stable.

Only environmental interpretation changes.

Canvas Across Devices

Different devices may implement Canvas differently.

Desktop.

Subtle texture.

Television.

Slightly deeper luminance.

Mobile.

Reduced texture for performance.

The perceived material should remain consistent regardless of implementation.

Canvas And Accessibility

Accessibility possesses higher authority than atmosphere.

The Canvas should always preserve:

- text readability

- contrast
- orientation

Atmospheric influence should automatically reduce whenever readability would otherwise decrease.

Canvas And Refraction

Canvas does not refract.

Instead...

It receives refracted light from surrounding Acrylic materials.

Conceptually.

[text]
Artwork

↓

Atmosphere

↓

Acrylic Refraction

↓

Canvas Reflection

This relationship subtly reinforces depth while preventing visual clutter.

Good Examples

Library

Large calm Canvas.

Artwork provides emotion.

Hero softly illuminated.

The environment feels spacious.

Playback

Canvas almost disappears.

Video dominates.

Overlay materials float naturally above it.

Nothing distracts from the entertainment.

Administration

Canvas becomes even calmer.

Atmosphere influence decreases.

The environment supports productivity without abandoning Mosaic identity.

Anti-patterns

Decorative Background

Canvas becomes artwork.

Content loses prominence.

Heavy Atmosphere

Canvas strongly adopts artwork colours.

The interface becomes visually noisy.

Flat Black

Large areas of featureless black.

Depth disappears.

The environment feels empty rather than calm.

Moving Background

Canvas animates independently from interaction.

Continuity weakens.

Canvas Model

[mermaid]
flowchart TD

```
NeutralFoundation
NeutralFoundation --> RuntimeAtmosphere
RuntimeAtmosphere --> Canvas
Canvas --> Composition
Composition --> Materials
Materials --> Presentation
```

The Canvas provides the environmental foundation upon which every Mosaic experience is built.

Relationship To Future Chapters

The following chapters build upon Canvas by introducing increasingly expressive materials.

Including:

- Acrylic

- Hero Material
- Overlay Material
- Refraction
- Light Transport

Canvas remains the calm foundation beneath them all.

Summary

Canvas is the quietest material within Mosaic.

It should rarely be noticed.

Instead it should create the feeling that:

- entertainment has room to breathe,
- materials belong somewhere,
- atmosphere exists naturally,
- and the user's World occupies one coherent physical environment.

When successful, users will remember the entertainment.

Not the surface beneath it.

Review Status

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04-acrylic.md

Acrylic

Purpose

Acrylic is the defining material of the Mosaic Design System.

It is the physical medium through which the user's entertainment influences the interface.

Unlike glass, Acrylic should not disappear.

Unlike opaque panels, Acrylic should not isolate itself from its surroundings.

Instead, Acrylic should behave like a premium physical material that:

- possesses depth,
- receives light,
- refracts colour,
- softly diffuses atmosphere,
- remains structurally present.

It is one of the most recognisable characteristics of the Mosaic visual language.

Definition

Within MDS, **Acrylic** is defined as:

A semi-translucent material that receives environmental light and Runtime Atmosphere while preserving hierarchy, readability and physical presence.

Acrylic is not:

- glass,
- blur,
- transparency,
- opacity,
- frosted panels.

Those are implementation techniques.

Acrylic is a material.

Philosophy

The industry often treats acrylic as:

Blur

+

Transparency

Mosaic intentionally rejects this.

Instead Acrylic behaves more like a solid object.

Imagine a thick polished acrylic tile placed onto a table.

It possesses:

- edges,
- thickness,
- internal diffusion,
- light transport,
- subtle reflections.

The interface should communicate that same feeling.

Acrylic Is Present

Glass attempts to disappear.

Acrylic should remain visible.

Not because it is visually loud...

but because it physically exists.

Users should perceive:

- depth,
- volume,
- softness,

rather than simply:

- blur.

This distinction creates a much more premium material language.

Acrylic Responsibilities

Acrylic performs five primary responsibilities.

1. Receive Atmosphere

Acrylic receives Runtime Atmosphere from surrounding entertainment.

It should never generate atmosphere independently.

2. Transport Light

Light should appear to travel through Acrylic.

This behaviour is formalised later through the Refraction System.

3. Preserve Hierarchy

Despite receiving atmospheric colour, Acrylic must remain structurally readable.

Hierarchy always takes precedence over visual richness.

4. Create Depth

Acrylic communicates physical layering.

It should feel thicker than ordinary interface panels.

5. Support Continuity

Atmosphere should move naturally through Acrylic during interaction.

Nothing should appear to flash or abruptly recolour.

Acrylic Is Not Transparent

Transparency implies seeing through a material.

Acrylic implies seeing **within** a material.

This distinction is important.

Poor.

Transparent Blur

↓

Background Visible

Preferred.

Light Enters

↓

Diffusion

↓

Refraction

↓

Soft Reflection

↓

Material

The material becomes believable rather than decorative.

Runtime Atmosphere

Runtime Atmosphere provides the energy.

Acrylic determines how that energy behaves.

Conceptually.

[text]
Artwork

↓

Runtime Atmosphere

↓

Acrylic

↓

Refraction

↓

Presentation

Acrylic should therefore be considered an interpreter of atmosphere rather than its source.

Diffusion

Incoming atmosphere should diffuse naturally.

Examples.

Strong artwork colours.

↓

Soft internal glow.

↓

Neutral readable surface.

The material should smooth environmental changes.

Not amplify them.

Refraction

Unlike ordinary translucent materials...

Acrylic should refract light.

Refraction should create:

- gentle colour movement,
- perceived depth,
- environmental richness.

It should never distort:

- typography,
- interaction,
- hierarchy.

Understanding remains more important than spectacle.

Internal Depth

Acrylic should possess perceived thickness.

Future implementations may communicate this through:

- layered diffusion,
- internal highlights,
- edge lighting,
- soft gradients.

Thickness should be perceived.

Not measured.

Edge Behaviour

Edges define Acrylic.

They communicate:

- structure,
- precision,
- craftsmanship.

Edges should respond more strongly to Runtime Atmosphere than central surfaces.

This creates the feeling that light is travelling through the material rather than simply colouring it.

Hero Acrylic

Hero Materials receive the strongest Acrylic treatment.

Examples include:

- stronger refraction,
- richer environmental light,
- greater perceived thickness.

Importantly...

The Hero should still remain secondary to the entertainment artwork itself.

Artwork remains the emotional centre.

Supporting Acrylic

Supporting materials should receive noticeably less atmospheric influence.

Examples include:

- timeline,
- progress,
- shelves,
- navigation.

Supporting Acrylic should feel related to the Hero without competing with it.

Overlay Acrylic

Overlay Acrylic prioritises readability.

Atmospheric influence should reduce significantly.

Interaction takes precedence over immersion.

Dialogs.

Menus.

Playback controls.

All require stronger separation from the environment.

Movement

Acrylic should respond naturally during interaction.

Changing Focus.

↓

Atmosphere gradually redistributes.

Moving Hero.

↓

Refraction subtly shifts.

Playback begins.

↓

Acrylic influence reduces.

Materials should appear physically consistent throughout every interaction.

Themes

Light Theme.

Acrylic appears:

- brighter,
- softer,
- more diffused.

Dark Theme.

Acrylic appears:

- deeper,
- richer,
- more luminous.

Both remain recognisably the same material.

Only environmental lighting changes.

Accessibility

Accessibility should always constrain Acrylic.

If diffusion or refraction would reduce:

- readability,
- contrast,
- orientation,

their intensity should reduce automatically.

Acrylic exists to enrich understanding.

Not complicate it.

Performance

Future implementations should optimise Acrylic aggressively.

Preferred techniques include:

- cached atmosphere,
- GPU acceleration,
- incremental updates,
- shared material layers.

Users should experience premium materials without perceiving computational cost.

Plugins

Extensions never define Acrylic.

Plugins contribute:

- Information,
- Artwork,
- Relationships.

The Material System determines how Acrylic behaves.

This guarantees every extension inherits the same physical language.

Good Examples

Hero

Poster softly illuminates Acrylic.

Edges gently glow.

Typography remains perfectly readable.

The material feels physically present.

Continue Watching

Timeline receives subtle atmospheric diffusion.

Hero remains dominant.

Nothing feels disconnected.

Playback

Controls float using restrained Acrylic.

Video remains visually dominant.

Interaction feels calm.

Anti-patterns

Frosted Glass

Heavy blur becomes the defining characteristic.

Material identity disappears.

Fully Transparent Panels

Background dominates foreground.

Depth weakens.

Plastic Shine

Strong specular highlights unrelated to Runtime Atmosphere.

The material feels artificial.

Decorative Refraction

Colour movement exists purely because it looks impressive.

No additional understanding is communicated.

Acrylic Model

[mermaid]
flowchart TD

```
RuntimeAtmosphere
RuntimeAtmosphere --> Acrylic
Acrylic --> Diffusion
Diffusion --> Refraction
Refraction --> MaterialDepth
MaterialDepth --> Presentation
```

Acrylic transforms environmental light into believable physical presence.

Relationship To Future Chapters

The following chapters expand Acrylic into increasingly sophisticated physical behaviour.

Including:

- Hero Material
- Overlay Material
- Refraction
- UV-Indexed Refraction
- Light Transport
- Runtime Material Resolution

These systems all build upon the Acrylic behaviour established here.

Summary

Acrylic is the defining material of Mosaic.

It should feel:

- substantial,
- refined,
- illuminated,
- calm.

It is not glass.

It is not blur.

It is the physical medium through which entertainment quietly reaches into the interface.

When successful, Acrylic should make the interface feel handcrafted rather than rendered.

Review Status

Status

Draft

Next File

05-hero-material.md

Hero Material

Purpose

The Hero Material is the most expressive material within the Mosaic Design System.

It represents the physical centre of the user's current World.

Unlike decorative hero banners found in traditional applications, the Hero Material exists to communicate:

- current Focus,
- current Context,
- current emotional atmosphere,
- compositional importance.

The Hero Material should feel as though it has been quietly illuminated by the entertainment itself.

It should never overshadow that entertainment.

Definition

Within MDS, **Hero Material** is defined as:

The highest-presence material within the current Composition, receiving the strongest interaction with Runtime Atmosphere while preserving readability, hierarchy and restraint.

Hero Material exists because something currently matters.

Not because every screen needs a visually impressive header.

Philosophy

The Hero Material should behave like a carefully machined sheet of premium acrylic positioned beneath the current entertainment.

Imagine placing a translucent acrylic panel beneath a beautifully printed film poster.

The acrylic receives:

- reflected colour,
- edge illumination,
- internal diffusion,

while the artwork remains sharp and emotionally dominant.

That relationship defines the Hero Material.

One Hero

Every Composition should normally contain exactly one Hero Material.

Poor.

Hero

Hero

Hero

Hero

Nothing leads.

Everything competes.

Preferred.

Hero

↓

Supporting Materials

↓

Canvas

The user's attention immediately knows where to begin.

Hero Owns Atmosphere

The Hero Material receives the strongest atmospheric influence of any material.

Conceptually.

[text]
Canvas

↓

Surface

↓

Acrylic

↓

Hero

Atmosphere weighting increases gradually.

The Hero should appear gently illuminated by the current artwork.

Never recoloured by it.

Hero Is Contextual

Hero Material should change with Focus.

Example.

Watching Frieren

↓

Hero

↓

Frieren Artwork

Changing Focus.

Watching Fire Force

↓

Hero Evolves

↓

Fire Force Artwork

The Hero Material evolves naturally.

The rest of the interface remains largely stable.

Hero Does Not Own Colour

The Hero Material receives colour.

It does not define colour.

Inputs include:

- Semantic Colours
- Runtime Atmosphere
- Theme
- Accessibility

The Hero Material is therefore an interpreter.

Not a source.

Hero Owns Depth

The Hero Material should possess the greatest perceived physical depth.

Characteristics include:

- strongest acrylic thickness,
- richest internal diffusion,
- deepest perceived volume,
- most convincing edge illumination.

Depth should be communicated through material behaviour.

Not exaggerated shadows.

Hero And Refraction

Hero Material is the primary consumer of the Refraction System.

Examples include:

- environmental colour transport,
- UV-indexed reflections,
- soft edge lighting,
- atmospheric diffusion.

Future rendering systems should prioritise Hero Material quality before secondary materials.

If performance budgets become constrained:

Supporting materials should simplify first.

Hero quality should remain as high as practical.

Hero And Artwork

Artwork always possesses higher authority than Hero Material.

Correct relationship.

[text]
Artwork

↓

Hero Material

↓

Supporting Interface

Incorrect.

[text]
Hero Material

↓

Artwork

The Hero should frame artwork.

Never compete with it.

Hero And Typography

Typography positioned upon Hero Material should always remain readable.

If Runtime Atmosphere reduces readability:

Atmosphere should reduce.

Typography should never compensate for material behaviour.

Understanding remains the highest priority.

Hero And Movement

Hero Material should evolve continuously.

Preferred.

[text]
Old Focus



Atmosphere Blends



Hero Evolves



New Focus

Avoid.

[text]
Old Hero



Removed



New Hero

The Hero should preserve identity wherever practical.

Users should feel continuity.

Not replacement.

Hero Across Themes

Light Theme.

Hero Material behaves like:

- softly illuminated acrylic,
- gentle diffusion,
- subtle edge light.

Dark Theme.

Hero Material behaves like:

- deeper acrylic,

- richer internal glow,
- stronger environmental lighting.

The material remains recognisably the same.

Only lighting interpretation changes.

Hero Across Devices

Desktop.

Large atmospheric acrylic.

Television.

Greater perceived depth.

Phone.

Reduced complexity.

Equivalent physical behaviour.

The Hero should remain conceptually identical across every supported device.

Hero And Playback

Playback intentionally reduces Hero prominence.

Before playback.

Hero supports exploration.

During playback.

Video becomes the Hero.

Overlay Materials replace Hero interaction.

After playback.

Hero gradually re-establishes itself around the next experience.

This transition should feel inevitable rather than mechanical.

Hero And Reading

Books behave differently.

Reading remains relatively static.

The Hero should therefore evolve more slowly.

Subtle atmospheric changes over long reading sessions reinforce calmness rather than cinematic energy.

Different domains may therefore produce different Hero rhythms while preserving identical material behaviour.

Hero And Accessibility

Accessibility should always constrain Hero Material.

Examples include:

- reduced glow,
- increased contrast,
- lower translucency,
- simplified refraction.

Hero identity should survive these adaptations.

The material should never become decorative at the expense of readability.

Performance

Hero Material should receive the highest rendering budget.

Future implementations should prioritise:

- shader quality,
- refraction quality,
- atmosphere precision,
- edge rendering.

Supporting materials may degrade gracefully under constrained hardware.

Hero quality should remain the primary visual investment.

Plugins

Extensions never create Hero Materials.

Plugins contribute:

- artwork,
- information,
- relationships.

The Composition Engine determines whether a Hero exists.

The Material System determines how that Hero behaves.

This guarantees one coherent visual identity throughout the ecosystem.

Good Examples

Film

Poster softly illuminates Hero Acrylic.

Edge lighting subtly follows artwork temperature.

Typography remains perfectly readable.

Users perceive atmosphere rather than effects.

Book

Illustrated cover casts warm reflected light into Hero Material.

Canvas remains calm.

Reading remains the primary activity.

Music

Album artwork generates subtle magenta environmental lighting.

Playback controls remain semantically coloured.

Artwork remains dominant.

Anti-patterns

Hero Wallpaper

Artwork becomes the interface background.

Hierarchy weakens.

Neon Hero

Atmosphere overwhelms typography.

The interface begins competing with entertainment.

Decorative Glow

Glow exists independently from Runtime Atmosphere.

Material behaviour becomes arbitrary.

Static Hero

Hero never changes despite Focus changing.

The interface becomes emotionally disconnected from the user's World.

Hero Material Model

[mermaid]
flowchart TD

```
HeroArtwork
HeroArtwork --> RuntimeAtmosphere
RuntimeAtmosphere --> HeroMaterial
HeroMaterial --> Refraction
Refraction --> Presentation
Presentation
Presentation --> CurrentFocus
```

Hero Material exists to strengthen the user's relationship with their current Focus.

It should never become the focus itself.

Relationship To Future Chapters

The following chapters build upon Hero Material by defining:

- Overlay Material
- Refraction
- UV-Indexed Refraction
- Light Transport
- Runtime Material Resolution

Hero Material represents the highest-presence physical object within the Material hierarchy.

The remaining materials derive much of their behaviour from it.

Summary

Hero Material is the emotional centre of the Mosaic interface.

It should feel:

- substantial,
- illuminated,
- refined,
- alive.

Not because it is visually extravagant...

but because it quietly reflects the user's current entertainment.

The Hero Material should create the feeling that the interface belongs inside the same world as the content without ever asking users to admire the interface itself.

Review Status

Status

Draft

Next File

06-overlay-material.md

Overlay Material

Purpose

Overlay Material represents temporary physical presence.

Unlike Canvas, which defines the environment...

or Hero Material, which expresses the user's current World...

Overlay Material exists only while interaction requires it.

Its responsibility is to provide clarity without breaking immersion.

When Overlay Material appears, users should feel that the interface has momentarily stepped forward to assist them.

Once its purpose has been fulfilled, it should quietly retreat back into the environment.

Definition

Within MDS, **Overlay Material** is defined as:

A temporary material layer that prioritises interaction, readability and focus while preserving continuity with the surrounding environment.

Overlay Material is intentionally transient.

It exists to support behaviour.

Not to permanently occupy the interface.

Philosophy

Imagine placing a polished acrylic sheet onto a desk while working.

It briefly becomes the object receiving your attention.

Once your work is complete...

It is removed.

The desk remains.

Overlay Material behaves the same way.

It temporarily becomes the physical centre of interaction before quietly disappearing again.

Temporary Presence

Overlay Material should never feel permanent.

Examples include:

- playback controls
- command palettes
- search
- dialogs
- menus
- confirmation sheets
- context menus

Every Overlay should communicate:

"I am here to help."

Never:

"I have replaced your World."

Overlay Owns Interaction

While active, Overlay Material temporarily receives the highest interaction priority.

However...

It should **not** receive the strongest atmospheric influence.

Interaction always possesses higher authority than immersion.

Conceptually.

Interaction

↓

Readability

↓

Atmosphere

This ordering should remain consistent across every Overlay.

Overlay And Atmosphere

Overlay Materials continue participating in Runtime Atmosphere.

However...

Atmospheric influence should be intentionally reduced.

Reasons include:

- readability
- contrast
- interaction precision

- accessibility

The Overlay should remain visually connected to the surrounding World without inheriting its full emotional intensity.

Overlay Is Acrylic

Overlay Material is derived from Acrylic.

It inherits:

- translucency
- diffusion
- refraction
- physical presence

It differs by modifying:

- atmosphere weighting
- contrast
- edge definition
- readability

Overlay Material is therefore a specialised Acrylic.

Not an independent material family.

Overlay And Canvas

Overlay Material should never disconnect from the Canvas.

Instead.

Canvas

↓

Overlay Emerges

↓

Canvas Remains Visible

↓

Overlay Leaves

↓

Canvas Continues

The environment remains continuous throughout the interaction.

Overlay And Hero

When Overlay Material appears above a Hero:

The Hero should quietly recede.

Not disappear.

Conceptually.

Hero

↓

Reduced Emphasis

↓

Overlay

↓

Interaction

↓

Overlay Closes

↓

Hero Returns

Users should never feel that the Hero has been replaced.

Only temporarily obscured.

Overlay And Playback

Playback provides a particularly important example.

Video becomes the Hero.

Overlay Material appears only while:

- seeking
- pausing
- adjusting subtitles
- changing audio
- browsing chapters

When controls disappear:

The video immediately regains visual dominance.

The Overlay never competes with playback longer than necessary.

Overlay And Search

Search is considered an Overlay interaction.

The user's current World remains beneath it.

Example.

Watching Frieren

↓

Search

↓

Results

↓

Close Search

↓

Watching Frieren

Focus has not necessarily changed.

The Overlay simply provided temporary assistance.

Overlay And Reading

Reading requires particularly restrained overlays.

Examples include:

- chapter navigation
- bookmarks
- glossary
- search

The Overlay should appear lightweight.

It should never interrupt reading rhythm unnecessarily.

Edge Behaviour

Overlay Materials should possess stronger edge definition than ordinary Acrylic.

This improves:

- separation
- focus
- readability

The effect should remain subtle.

Users should perceive confidence rather than borders.

Refraction

Overlay Material participates in Refraction.

However...

Refraction intensity should reduce.

Reasons include:

- readability
- interaction precision
- accessibility

The Overlay should still feel physically related to surrounding Acrylic.

It simply behaves more conservatively.

Motion

Overlay Material should feel physically connected to the World.

Preferred.

over lay

↓

Emerges

↓

Interaction

↓

Returns

Avoid.

over lay

↓

Appears

↓

Disappears

Movement should communicate physical presence.

Not interface visibility.

Accessibility

Overlay Material must always prioritise:

- readability
- contrast
- orientation
- focus

If Runtime Atmosphere would reduce interaction quality:

Its influence should automatically decrease.

Accessibility remains the highest authority.

Performance

Overlay Materials should favour stability over complexity.

Preferred.

- cached atmosphere
- simplified refraction
- restrained diffusion

Interaction should always remain highly responsive.

Users should never perceive latency introduced by material rendering.

Plugins

Extensions should never implement Overlay Materials independently.

Plugins request interactions.

The platform presents them.

Every Overlay should therefore inherit:

- Material behaviour
- Runtime Atmosphere
- Accessibility
- Motion

automatically.

Good Examples

Playback Controls

Controls softly emerge from the surrounding Acrylic.

Video remains recognisable beneath.

Interaction feels immediate.

Search

Search appears as a confident material layer.

The previous Composition remains visible beneath it.

Closing Search naturally restores the previous Context.

Context Menu

Menu briefly becomes the interaction centre.

Atmosphere remains subtle.

Typography remains perfectly readable.

Anti-patterns

Opaque Dialogs

Overlay completely disconnects from the surrounding World.

Continuity weakens.

Decorative Blur

Heavy blur dominates interaction.

Readability decreases.

Floating Glass

Overlay appears unrelated to the surrounding Material System.

Physical coherence disappears.

Permanent Overlay

Temporary materials remain visible after interaction ends.

The interface begins feeling cluttered.

Overlay Material Model

[mermaid]
flowchart TD

```
Canvas
Canvas --> Hero
Hero --> Overlay
Overlay --> Interaction
Interaction --> OverlayDismissed
OverlayDismissed --> Hero
Hero --> Canvas
```

Overlay temporarily becomes the centre of interaction.

The World beneath it never disappears.

Relationship To Future Chapters

The following chapters expand upon the physical behaviour shared by every material.

Including:

- Refraction
- UV-Indexed Refraction
- Light Transport
- Runtime Material Resolution

Overlay Material demonstrates how those systems should adapt when interaction becomes more important than atmosphere.

Summary

Overlay Material exists to help.

It should appear:

- confident,
- readable,
- physically believable,
- temporary.

It should never interrupt the user's relationship with their entertainment.

Instead, it should quietly step forward when needed...

...and quietly disappear once its work is complete.

That restraint is what makes Overlay Material feel like part of Mosaic rather than another layer placed on top of it.

Review Status

Status

Draft

Next File

07-refraction.md

Refraction

Purpose

Refraction is the defining physical behaviour of the Mosaic Material System.

It is the mechanism through which entertainment subtly influences the interface.

Without Refraction, Acrylic becomes translucent decoration.

With Refraction, Acrylic becomes a believable physical material.

Refraction transforms Runtime Atmosphere into the feeling that light is travelling through the interface itself.

It is one of the primary visual signatures of Mosaic.

Definition

Within MDS, **Refraction** is defined as:

The controlled transport and diffusion of environmental light through Mosaic materials in order to reinforce immersion, hierarchy and physical presence.

Refraction is not:

- blur
- bloom
- colour overlays
- gradients
- transparency

Those are rendering techniques.

Refraction is a physical behaviour.

Philosophy

Imagine a polished sheet of premium acrylic placed beside a colourful book cover.

The acrylic does not become the same colour as the artwork.

Instead...

Light enters the acrylic.

Travels through it.

Softly bends.

Diffuses internally.

Then exits again.

This behaviour creates the impression of physical depth.

That is the behaviour Mosaic seeks to reproduce.

Why Refraction Exists

Entertainment already contains:

- colour
- light
- atmosphere
- emotional tone

Traditional interfaces isolate themselves from these qualities.

Mosaic instead allows those qualities to gently influence surrounding materials.

Refraction creates the feeling that:

The interface exists inside the same environment as the entertainment.

Light Transport

Refraction should always be thought of as light transport.

Not colour replacement.

Poor.

[text]
Artwork

↓

Copy Colours

↓

Interface

Preferred.

[text]
Artwork

↓

Light

↓

Material

↓

Refraction

↓

Interface

The distinction is critical.

Materials remain physically believable.

Brand identity remains intact.

Refraction Inputs

Refraction receives several conceptual inputs.

[text]

Runtime Atmosphere

↓

Material Type

↓

Material Thickness

↓

Surface Orientation

↓

Composition Importance

Refraction should never depend directly upon component implementation.

It is a material behaviour.

Refraction Outputs

Refraction influences:

- edge lighting
- internal diffusion
- subtle colour transport
- environmental glow
- perceived depth

It should never directly influence:

- typography
- icons
- interaction affordances

Understanding remains independent from refraction.

Refraction Strength

Different materials receive different refraction intensity.

Material	Refraction
Canvas	None
Surface	Very Low
Acrylic	Medium
Hero	High
Overlay	Low

This hierarchy mirrors the Material Hierarchy.

Refraction reinforces physical importance.

It does not define it.

Directionality

Refraction should possess direction.

Light should appear to originate from:

Hero Artwork

rather than uniformly illuminating every surface.

Conceptually.

[text]
Artwork

↓

Light Direction

↓

Nearby Materials

↓

Environmental Falloff

This creates a much stronger sense of physical coherence.

Diffusion

Light should soften as it travels through materials.

Strong artwork colours should gradually become:

- calmer
- broader
- less saturated

Diffusion prevents the interface from becoming visually noisy.

Users should perceive atmosphere rather than individual colours.

Edge Refraction

Edges should refract more strongly than flat surfaces.

Reasons include:

- perceived thickness
- physical realism
- hierarchy

Edge lighting should remain subtle.

Users should feel depth.

Not notice glowing borders.

Hero Refraction

Hero Material receives the highest quality refraction.

Examples include:

- richer colour transport
- deeper diffusion
- stronger perceived volume
- smoother temporal blending

Hero Refraction should remain emotionally expressive without becoming visually dominant.

The artwork always remains sharper and more detailed than the surrounding material.

Supporting Refraction

Supporting Acrylic receives restrained refraction.

Examples include:

- shelves
- timelines
- continuation panels

These materials should visually relate to the Hero without competing for attention.

Overlay Refraction

Overlay Material intentionally reduces refraction.

Interaction possesses higher priority than environmental realism.

Menus.

Playback controls.

Dialogs.

Search.

All should preserve clarity before atmosphere.

Temporal Behaviour

Refraction should evolve continuously.

Preferred.

[text]
Artwork Changes

↓

Light Shifts

↓

Refraction Evolves

↓

Materials Settle

Avoid.

[text]
Artwork Changes

↓

Immediate Recolour

The interface should appear physically illuminated rather than digitally updated.

Composition Awareness

Refraction should respect Composition.

Primary materials.

↓

Stronger environmental response.

Peripheral materials.

↓

Minimal environmental response.

Composition therefore influences physical behaviour without components becoming aware of it.

Accessibility

Refraction should never reduce:

- readability
- contrast
- orientation

If environmental light conflicts with semantic clarity:

Refraction should reduce automatically.

Accessibility always overrides visual richness.

Performance

Future implementations should optimise refraction aggressively.

Recommended conceptual strategies include:

- cached atmosphere
- simplified edge calculations
- temporal interpolation
- GPU acceleration
- shared material buffers

Users should experience premium materials without perceiving rendering cost.

Plugins

Extensions never participate in refraction.

Plugins contribute:

- artwork
- information

The Material System determines:

- light transport
- diffusion
- edge behaviour

Every extension therefore inherits identical material quality automatically.

Good Examples

Science Fiction

Cool artwork softly illuminates Hero Acrylic.

Nearby materials receive subtle blue reflections.

Typography remains perfectly readable.

Fantasy Novel

Warm golden artwork creates soft amber edge lighting.

Canvas remains neutral.

Atmosphere feels warm rather than colourful.

Music

Album artwork introduces subtle magenta highlights into nearby Acrylic.

Playback controls remain restrained.

Attention stays with the album cover.

Anti-patterns

Colour Overlay

Artwork colours directly tint components.

Materials lose physical credibility.

Neon Refraction

Highly saturated glowing edges.

Atmosphere becomes visual spectacle.

Uniform Lighting

Every material receives identical illumination.

Hierarchy disappears.

Animated Refraction

Refraction continually shifts without behavioural cause.

Users become distracted.

Refraction Model

[mermaid]
flowchart TD

```
Artwork
Artwork --> RuntimeAtmosphere
RuntimeAtmosphere --> Material
Material --> Refraction
Refraction --> Diffusion
Diffusion --> Presentation
Presentation
Presentation --> UserPerception
```

Refraction transforms environmental light into believable physical materials.

Relationship To Future Chapters

The following chapter expands this concept through the **UV-Indexed Refraction System**.

Rather than treating materials as uniformly illuminated, Mosaic introduces spatial light transport based upon UV coordinates.

This allows atmosphere to:

- move naturally,
- preserve hierarchy,
- remain computationally efficient,
- closely follow the user's current Hero.

It is expected to become one of the defining technical innovations of the Mosaic Material System.

Summary

Refraction is the mechanism through which the user's entertainment quietly reaches into the interface.

It should feel:

- physical,
- restrained,
- believable,
- continuous.

Users should never think:

"That shader looks impressive."

Instead they should simply feel that the interface belongs beside the entertainment.

That subtle shift in perception is the ultimate purpose of Refraction.

Review Status

Status

Draft

Next File

08-uv-indexed-refraction.md

UV-Indexed Refraction

Purpose

The previous chapter established **Refraction** as the physical behaviour through which Runtime Atmosphere travels through Mosaic materials.

This chapter introduces the mechanism that makes that behaviour practical.

UV-Indexed Refraction.

Rather than treating every material as receiving uniform environmental lighting, Mosaic maps atmospheric influence into a normalised spatial coordinate system.

This allows Runtime Atmosphere to behave like physical light rather than decorative colour.

UV-Indexed Refraction is expected to become one of the defining technical characteristics of the Mosaic Material System.

Philosophy

Entertainment should illuminate the interface naturally.

Not uniformly.

Imagine sunlight entering a room.

Surfaces closest to the window receive stronger light.

More distant surfaces receive less.

The same principle applies within Mosaic.

The Hero acts as the conceptual light source.

Every surrounding material receives atmosphere according to its relationship with that Hero.

Not according to arbitrary layout.

Definition

Within MDS, **UV-Indexed Refraction** is defined as:

A spatial lighting model in which Runtime Atmosphere is projected into a normalised coordinate system and sampled by materials according to their relationship with the current Hero.

The system describes:

- where light exists,
- how it propagates,
- how materials sample it,

independently from rendering technology.

Why UV Coordinates?

Screen coordinates are unstable.

Examples.

Desktop.

1920 × 1080

Tablet.

2048 × 1536

Phone.

1179 × 2556

Using pixels would require every client to solve atmosphere differently.

Instead Mosaic normalises every composition into UV space.

0.0 → 1.0

Both axes.

Every client therefore shares one conceptual lighting model.

Presentation adapts.

The lighting model remains identical.

Conceptual UV Space

Every Composition generates a normalised atmospheric space.

```
[text]
0,0 ----- 1,0
|
|
|
|
0,1 ----- 1,1
```

The Hero defines the primary light origin.

Runtime Atmosphere is projected into this coordinate space.

Materials sample from it.

Hero As Light Source

The Hero does not emit colour.

It emits environmental influence.

Conceptually.

```
[text]
Hero Artwork
```

↓

Colour Extraction

↓

Runtime Atmosphere

↓

UV Light Field

Every surrounding material samples from this shared environmental field.

This creates one coherent visual environment.

Light Falloff

Environmental influence should decrease naturally with conceptual distance.

Example.

[text]
Hero

↓

Timeline

↓

Related Works

↓

Peripheral Collections

The Timeline should receive stronger atmospheric influence than Peripheral Collections.

This is **not** because of pixels.

It is because of compositional relationship.

The UV field therefore reflects the Composition rather than the screen.

Conceptual Rather Than Geometric

The UV field should follow conceptual space.

Not physical space.

Example.

A Hero positioned at the bottom of a television interface should still illuminate conceptually related surfaces.

The implementation may project that field differently.

The conceptual relationship remains unchanged.

This distinction allows Composition to remain device independent.

Material Sampling

Every material samples the UV field differently.

Material	Sampling Strength
Canvas	Minimal
Surface	Low
Acrylic	Medium
Hero	Maximum
Overlay	Reduced

Sampling strength reflects Material Hierarchy.

Not rendering capability.

Multi-Layer Refraction

The UV field should contain multiple conceptual channels.

Examples include:

[text]
Atmosphere

↓

Hue

↓

Intensity

↓

Temperature

↓

Direction

Materials may sample these independently.

Future renderers remain free to implement these channels differently while preserving identical behaviour.

Directional Lighting

The UV field should preserve directional information.

Example.

[text]
Hero

↓

Upper Left

↓

Light propagates diagonally

Materials therefore know not only:

- how much atmosphere exists,

but also:

- where it originates.

This dramatically improves perceived physicality.

Dynamic Reprojection

When Focus changes:

The UV field should not reset.

Preferred.

[text]
Old Hero

↓

Blend

↓

Field Reprojects

↓

New Hero

Avoid.

[text]
Old Field

↓

Delete

↓

New Field

Users should perceive environmental evolution.

Not regeneration.

Domain Independence

The UV model should remain identical across every domain.

Television.

↓

Poster.

Books.

↓

Cover.

Music.

↓

Album Artwork.

Games.

↓

Key Art.

Different media.

One lighting model.

Performance Strategy

The UV field should be generated once per meaningful atmospheric change.

Typical invalidation events include:

- Hero changes
- artwork changes
- theme changes
- accessibility changes

Ordinary interaction should sample the existing field rather than regenerate it.

Future implementations are expected to cache UV fields aggressively.

Resolution Independence

Because UV coordinates are normalised:

Desktop.

↓

High resolution.

Phone.

↓

Lower resolution.

Television.

↓

4K.

The lighting model remains identical.

Only sampling precision changes.

Refraction Sampling

Conceptually.

[text]
Runtime Atmosphere

↓

UV Field

↓

Material Samples

↓

Diffusion

↓

Presentation

Every material samples from the same atmospheric environment.

This creates coherence throughout the interface.

Accessibility

Accessibility should constrain UV influence.

If atmospheric gradients reduce:

- contrast,
- readability,
- focus,

sampling strength should automatically decrease.

The UV model should preserve understanding before realism.

Relationship To Motion

The Motion System should never animate UV coordinates directly.

Instead.

Behaviour changes.

↓

Composition changes.

↓

UV field updates.



Materials naturally respond.

The physical world appears to evolve because understanding evolved.

Not because an animation was added.

Future Implementation

Although implementation belongs to future specifications, the conceptual architecture supports technologies such as:

- GPU shaders
- Skia Runtime Effects
- Metal fragment pipelines
- Vulkan shaders
- WebGPU
- WebGL
- Flutter Impeller

All implementations should preserve the same conceptual UV behaviour.

Rendering technology remains an implementation detail.

Plugins

Extensions never interact with UV space.

Plugins contribute:

- artwork
- relationships
- information

The platform constructs:

- Runtime Atmosphere
- UV field
- Material behaviour

This guarantees one consistent environmental model across every extension.

Good Examples

Film

Poster becomes Hero.



Cool environmental light.



Nearby Acrylic samples stronger.



Peripheral materials sample less.



The interface feels physically illuminated.

Book

Illustrated cover.



Warm atmosphere.



Hero edges glow softly.



Reading controls remain neutral.

The environment supports reading rather than distracting from it.

Music

Album artwork.



Subtle magenta atmosphere.



Playback controls remain readable.



Hero receives strongest response.

The user's attention remains on the album.

Anti-patterns

Screen Space Lighting

Lighting determined entirely by screen coordinates.

Composition is ignored.

Uniform Atmosphere

Every surface receives identical atmospheric influence.

Hierarchy weakens.

Per-Component Lighting

Each component independently calculates atmosphere.

The interface fragments.

Colour Projection

Artwork colours copied directly onto interface surfaces.

Physical credibility disappears.

UV Refraction Model

[mermaid]
flowchart TD

```
HeroArtwork
HeroArtwork --> RuntimeAtmosphere
RuntimeAtmosphere --> UVField
UVField
UVField --> Canvas
UVField
UVField --> Surface
UVField
UVField --> Acrylic
UVField
UVField --> Hero
UVField
UVField --> Overlay
Canvas --> Presentation
Surface --> Presentation
Acrylic --> Presentation
Hero --> Presentation
Overlay --> Presentation
```

One environmental field.

Many materials.

One coherent world.

Relationship To Future Chapters

The next chapter defines **Light Transport**.

Where UV-Indexed Refraction explains:

Where light exists

Light Transport explains:

How that light physically moves through the Material System.

Together these chapters establish the physical simulation model that distinguishes Mosaic from traditional interface frameworks.

Summary

UV-Indexed Refraction transforms Runtime Atmosphere from a colour system into an environmental lighting model.

Rather than colouring components...

Mosaic creates one shared atmospheric field that every material samples according to its role within the Composition.

This produces an interface that feels:

- coherent,
- physical,
- calm,
- immersive.

Users should feel as though their entertainment is gently illuminating the world around it.

Not recolouring the software beneath it.

Review Status

Status

Draft

Next File

09-light-transport.md

Light Transport

Purpose

The previous chapters established:

- Runtime Atmosphere
- Acrylic
- Refraction
- UV-Indexed Refraction

This chapter defines the conceptual behaviour that connects them.

Light Transport describes how environmental light moves through the Mosaic Material System.

Unlike traditional user interfaces, which simply paint colour onto surfaces, Mosaic models the movement of light as a first-class architectural concept.

The objective is not physical simulation.

The objective is perceived physicality.

Definition

Within MDS, **Light Transport** is defined as:

The conceptual movement of environmental light through the Mosaic Material System in order to communicate depth, hierarchy and atmosphere.

Light Transport is an architectural model.

Not a rendering algorithm.

Philosophy

Imagine entering a modern gallery.

Light enters the room.

It reflects from paintings.

Travels across polished acrylic.

Softly illuminates nearby surfaces.

Nothing glows independently.

Everything participates in the same environment.

That is the experience Mosaic attempts to recreate.

The interface should feel illuminated.

Not coloured.

Light Is Shared

One of the most important rules within Mosaic is:

There is only one environment.

Every material participates in that environment.

Poor.

[text]
Hero

↓

Own Lighting

Preferred.

[text]
Environment

↓

Shared Light

↓

Many Materials

The Material System should feel physically coherent.

Light Origin

Environmental light originates conceptually from:

[text]
Current Hero

↓

Runtime Atmosphere

↓

Environmental Light

The Hero does not emit visible light.

Instead it influences the environmental lighting field.

The distinction prevents exaggerated visual effects while preserving immersion.

Transport Stages

Conceptually, light moves through several stages.

[text]
Artwork

↓

Atmosphere

↓

UV Field

↓

Material Sampling

↓

Diffusion

↓

Refraction

↓

Presentation

Each stage contributes one responsibility.

No stage duplicates another.

Material Response

Different materials transport light differently.

Material	Behaviour
Canvas	Receives ambient light
Surface	Receives soft diffusion
Acrylic	Refracts and diffuses
Hero	Highest environmental response
Overlay	Controlled, readability-first response

This hierarchy mirrors the Material Hierarchy established earlier.

Energy Conservation

Light Transport should preserve perceived energy.

If one material receives stronger atmospheric influence...

Nearby materials should naturally receive less.

The environment should feel balanced.

Not uniformly illuminated.

This principle prevents visual overload while improving perceived realism.

Diffusion

Transported light should soften naturally.

Strong local colour should gradually become:

- broader
- calmer
- less saturated

Diffusion prevents materials from feeling digitally coloured.

Instead they appear physically illuminated.

Edge Transport

Edges should transport light differently from flat surfaces.

Conceptually.

[text]
Incoming Light

↓

Internal Material

↓

Edge Highlight

↓

Soft Exit

Edges communicate:

- thickness
- craftsmanship
- physical presence

Future renderers may implement this differently.

The conceptual behaviour should remain stable.

Hero Influence

Hero Material possesses the strongest transport behaviour.

Nearby Acrylic should inherit:

- colour temperature
- ambient luminance
- subtle reflected highlights

Peripheral materials should inherit significantly less.

The Hero therefore becomes the environmental centre without becoming visually dominant.

Temporal Transport

Light should evolve continuously.

Preferred.

[text]
Artwork Changes

↓

Atmosphere Blends

↓

Light Redistributes

↓

Materials Respond

Avoid.

[text]
Artwork Changes

↓

Entire Interface Recolours

Users should perceive environmental continuity rather than colour replacement.

Composition Awareness

Light Transport should respect Composition.

Primary concepts receive stronger environmental participation.

Peripheral concepts remain calmer.

Composition therefore influences perceived physicality without components requiring knowledge of lighting behaviour.

Device Independence

The conceptual transport model should remain identical across:

- Desktop
- Television
- Mobile
- Tablet

Different devices may implement:

- sampling precision
- shader complexity
- blur quality

The perceived physical behaviour should remain recognisably Mosaic.

Accessibility

Accessibility constrains Light Transport.

Examples.

High Contrast.

↓

Reduced diffusion.

Reduced Motion.

↓

Simplified temporal blending.

Low Vision.

↓

Reduced atmospheric variation.

The Material System should always preserve readability before physical realism.

Performance Strategy

Future implementations should optimise Light Transport through:

- cached atmospheric fields
- incremental updates
- GPU acceleration
- temporal interpolation
- shared material buffers

Light Transport should never require complete recomputation during ordinary interaction.

Only meaningful environmental changes should trigger recalculation.

Runtime Updates

Typical recalculation events include:

- Hero changes
- Focus changes

- artwork changes
- theme changes
- accessibility changes

Scrolling, hovering or small UI interactions should generally reuse existing transport data.

This keeps the interface visually stable while remaining computationally efficient.

Plugin Participation

Extensions contribute:

- artwork
- metadata
- information

Plugins never participate directly in Light Transport.

The platform constructs one shared lighting model for every extension.

This guarantees a consistent physical language throughout the ecosystem.

Good Examples

Film

Poster introduces cool environmental light.

Hero Acrylic receives strongest transport.

Nearby tiles inherit subtle reflected lighting.

Canvas remains calm.

Book

Illustrated cover creates warm environmental lighting.

Reading controls remain largely neutral.

Atmosphere supports reading without distraction.

Music

Album artwork softly influences playback surfaces.

Controls remain semantically coloured.

The interface feels unified.

Anti-patterns

Independent Lighting

Every component calculates its own environmental response.

The interface fragments.

Static Lighting

Materials ignore changing artwork.

The interface feels disconnected from entertainment.

Decorative Bloom

Light exists because it looks dramatic.

No additional understanding is created.

Colour Flood

Environmental light saturates every surface equally.

Hierarchy disappears.

Light Transport Model

[mermaid]
flowchart TD

```
HeroArtwork
HeroArtwork --> RuntimeAtmosphere
RuntimeAtmosphere --> UVField
UVField --> LightTransport
LightTransport
LightTransport --> Canvas
LightTransport
LightTransport --> Surface
LightTransport
LightTransport --> Acrylic
LightTransport
LightTransport --> Hero
LightTransport
LightTransport --> Overlay
Hero
Hero --> Presentation
Acrylic
Acrylic --> Presentation
Surface
Surface --> Presentation
Canvas
Canvas --> Presentation
Overlay
Overlay --> Presentation
```

One shared environmental light model.

Many material responses.

One coherent physical world.

Relationship To Future Chapter

The next chapter defines **Runtime Material Resolution**.

Where Light Transport explains:

How environmental light moves

Runtime Material Resolution explains:

How every material resolves into concrete runtime behaviour.

It is the bridge between conceptual material behaviour and implementation.

Summary

Light Transport is the physical behaviour that unifies every material within Mosaic.

Rather than colouring individual components, the platform creates one coherent environmental lighting model that every material interprets according to its role.

The result should feel:

- physical,
- calm,
- believable,
- emotionally connected.

Users should never think about light.

They should simply feel that the interface naturally belongs beside the entertainment they love.

Review Status

Status

Draft

Next File

10-runtime-material-resolution.md

Runtime Material Resolution

Purpose

The previous chapters established the conceptual behaviour of the Material System.

They defined:

- Material Hierarchy
- Acrylic
- Hero Material
- Overlay Material
- Refraction
- UV-Indexed Refraction
- Light Transport

This chapter defines how those independent systems become a resolved material at runtime.

Runtime Material Resolution is the bridge between architectural intent and rendered surfaces.

It ensures that every Mosaic client presents identical material behaviour regardless of rendering technology.

Definition

Within MDS, **Runtime Material Resolution** is defined as:

The deterministic process through which conceptual material behaviour is transformed into concrete runtime material properties suitable for rendering.

Runtime Material Resolution never changes material identity.

It only determines how that identity is expressed in the current environment.

Why Resolution Exists

A Hero Tile should never decide:

- blur strength
- acrylic depth
- edge lighting
- atmosphere intensity
- translucency
- refraction strength

Instead it simply requests:

[text]

Material.Hero

The Runtime Material Resolver determines everything else.

This dramatically reduces component complexity.

Resolution Pipeline

Every material follows the same conceptual pipeline.

[text]
Material Identity

↓

Semantic Tokens

↓

Runtime Atmosphere

↓

UV Field

↓

Light Transport

↓

Accessibility

↓

Device Capability

↓

Resolved Material

Each stage contributes exactly one responsibility.

Resolution Inputs

The Runtime Material Resolver evaluates:

[text]
Current World

↓

Current Focus

↓

Current Context

↓

Composition

↓

Material Identity

↓

Runtime Atmosphere

↓

Accessibility

↓

Theme

↓

Device

No single input should dominate.

The resolver balances all inputs according to architectural priority.

Resolution Priority

Material Resolution follows a strict evaluation order.

[text]

1.

Material Identity

↓

2.

Composition

↓

3.

Runtime Atmosphere

↓

4.

Accessibility

↓

5.

Device Capability

↓

6.

Rendering Backend

Meaning always precedes implementation.

Accessibility always precedes aesthetics.

Material Identity Never Changes

One of the strongest guarantees within Mosaic is:

[text]
Hero Material

↓

Always Hero Material

Regardless of:

- device
- theme
- accessibility
- artwork

Only its physical implementation changes.

The conceptual identity remains constant.

Runtime Adaptation

Runtime Material Resolution adapts implementation.

Examples include:

Hero.

↓

Greater perceived thickness.

Playback.

↓

Reduced atmosphere.

Reading.

↓

Softer diffusion.

Administration.

↓

Calmer materials.

The same Material Identity behaves differently because the user's World changed.

Not because components changed.

Accessibility Resolution

Accessibility possesses higher authority than physical realism.

Examples.

High Contrast.



Reduced translucency.

Reduced Motion.



Simplified atmosphere interpolation.

Low Vision.



Reduced refraction.

Every adaptation should preserve:

- hierarchy
- readability
- interaction

before preserving material richness.

Device Resolution

Different devices possess different rendering capabilities.

Desktop.



Full Acrylic.

Television.



Enhanced depth.

Phone.



Reduced shader complexity.

Low Power Device.



Simplified diffusion.

Despite implementation differences...

Users should continue perceiving the same Material System.

Material Profiles

Future implementations may internally generate Material Profiles.

Conceptually.

[text]
Material.Hero

↓

Runtime Profile

↓

Blur

↓

Refraction

↓

Thickness

↓

Edge Behaviour

↓

Lighting

↓

Resolved Material

Components consume only the completed profile.

Runtime Caching

Resolved Materials should be aggressively cached.

Typical cache invalidation events include:

- Hero changes
- Focus changes
- artwork changes
- theme changes
- accessibility changes

Ordinary scrolling should not invalidate material resolution.

The environment should remain visually stable.

Incremental Updates

Material Resolution should favour incremental refinement.

Preferred.

[text]
Atmosphere Changes

↓

Update Hero

↓

Update Nearby Acrylic

↓

Canvas Stable

Avoid.

[text]
Atmosphere Changes

↓

Rebuild Entire Material Tree

Incremental updates preserve continuity while reducing computational cost.

Composition Awareness

Material Resolution should remain Composition-aware.

Example.

Primary Composition.

↓

High-quality Acrylic.

Peripheral Composition.

↓

Simplified Acrylic.

Material fidelity should follow compositional importance.

Rendering effort should therefore support user understanding.

Plugin Behaviour

Extensions never resolve materials.

Plugins contribute:

- artwork

- information
- relationships

The platform resolves:

- Material Identity
- Atmosphere
- Refraction
- Lighting

This guarantees every extension inherits identical physical behaviour.

Good Examples

Hero

Current artwork.



Atmosphere.



Hero Profile.



Premium Acrylic.



Rendered Surface.

Timeline

Supporting Composition.



Supporting Acrylic.



Reduced refraction.



Consistent hierarchy.

Playback Overlay

Overlay Material.



Reduced atmosphere.

↓

Maximum readability.

↓

Interaction remains effortless.

Anti-patterns

Component Materials

Components constructing materials independently.

The Material System fragments.

Device Materials

Every client inventing its own material hierarchy.

Consistency disappears.

Runtime Identity

Runtime changing Hero into Surface.

Meaning has leaked into implementation.

Complete Rebuild

Every runtime update regenerates every material.

Continuity weakens.

Performance decreases.

Runtime Material Model

[mermaid]
flowchart TD

```
MaterialIdentity  
MaterialIdentity --> Composition  
Composition --> RuntimeAtmosphere  
RuntimeAtmosphere --> LightTransport  
LightTransport --> Accessibility  
Accessibility --> DeviceCapability  
DeviceCapability --> ResolvedMaterial  
ResolvedMaterial --> Presentation
```

Material behaviour is resolved once.

Components simply consume the result.

Relationship To Future Specifications

Future specifications are expected to formalise:

- Material Resolver
- GPU Material Pipeline
- Shader Profiles
- Material Cache
- Acrylic Renderer
- Refraction Engine
- Cross-platform Material Backends

These systems implement the architecture defined by this chapter.

Summary

Runtime Material Resolution transforms conceptual materials into physical experience.

It preserves:

- hierarchy
- atmosphere
- accessibility
- continuity
- performance

while hiding implementation complexity from the rest of the platform.

Components should never know:

- how Acrylic is rendered,
- how light was transported,
- how atmosphere was generated.

They should simply receive:

Material.Hero

The Material System does everything else.

Review Status

Status

Draft

Next File

Material System Governance

Purpose

The Material System defines the physical language of Mosaic.

Unlike implementation techniques, materials become part of the user's long-term perception of the product.

People may never consciously remember:

- blur radius,
- shader implementation,
- translucency algorithms.

They will remember:

- how substantial the interface felt,
- whether it felt calm,
- whether it felt premium,
- whether it belonged beside their entertainment.

This chapter defines how the Material System should evolve while preserving that identity.

Governance Philosophy

Materials should evolve technologically.

They should remain behaviourally consistent.

The objective is not preserving shaders.

It is preserving perception.

Future rendering engines may completely replace today's implementation.

Users should still recognise:

This feels like Mosaic.

Material Identity

Every Material within Mosaic possesses an architectural identity.

Examples include:

- Canvas
- Surface
- Acrylic
- Hero

- Overlay

These identities are considered part of the public Design System.

Changing them requires architectural review.

Changing their implementation generally does not.

Stable Responsibilities

The following concepts should remain highly stable.

- Material hierarchy
- Acrylic philosophy
- Hero behaviour
- Overlay behaviour
- Refraction model
- Light transport model

These concepts define the physical identity of Mosaic.

Evolvable Responsibilities

The following may evolve over time.

- shader implementations
- blur algorithms
- rendering pipelines
- GPU techniques
- performance optimisations
- platform-specific implementations

These improvements should preserve identical perceived behaviour.

Material Ownership

Material responsibilities are intentionally separated.

Layer	Owner
Material Philosophy	Design Systems
Material Hierarchy	Design Systems
Runtime Material Resolution	Runtime Platform
Rendering Backend	Client Platform
GPU Implementation	Rendering Layer

Ownership preserves architectural integrity while allowing implementation to evolve independently.

Introducing New Materials

Before introducing a new Material ask:

Question One

Can an existing material already express this behaviour?

Question Two

Is this genuinely a new physical material...

or merely another implementation?

Question Three

Could Runtime Material Resolution solve this instead?

Question Four

Will users perceive this as a distinct physical object?

Question Five

Will this material remain meaningful independently from today's rendering technology?

If uncertainty remains...

The proposal should be refined before implementation.

Material Drift

Material Drift occurs when:

- similar materials behave differently,
- rendering implementations diverge,
- new effects bypass the Material System,
- platforms invent independent behaviours.

Material Drift weakens the physical language of Mosaic.

Over time the interface begins feeling assembled rather than designed.

Material Drift should therefore be treated as architectural debt.

Acrylic Governance

Acrylic represents one of the defining visual characteristics of Mosaic.

Changes affecting Acrylic require particular care.

Examples include:

- translucency
- diffusion
- edge behaviour
- refraction
- perceived thickness

Future improvements should strengthen the illusion of physical presence rather than chasing visual novelty.

Refraction Governance

Refraction should always remain:

- subtle,
- purposeful,
- behaviourally justified.

Refraction should never become:

- decorative,
- animated for its own sake,
- independent from Runtime Atmosphere.

Users should feel environmental lighting.

Not visual effects.

Runtime Governance

Runtime Material Resolution may evolve frequently.

Examples include:

- better GPU techniques,
- improved caching,
- adaptive sampling,
- HDR support,
- platform optimisation.

These improvements should require no changes to:

- Components
- Composition
- Semantic Tokens

The Material System should absorb complexity.

Applications should remain simple.

Accessibility Governance

Accessibility possesses higher authority than Material fidelity.

If Material behaviour reduces:

- readability,
- orientation,
- interaction,

Material behaviour should adapt automatically.

No Material proposal should weaken accessibility in pursuit of realism.

Plugin Governance

Extensions must never introduce:

- custom materials,
- independent acrylic systems,
- proprietary refraction,
- separate physical hierarchies.

Plugins inherit the Material System.

They never redefine it.

This ensures one coherent physical language across the entire Mosaic ecosystem.

Review Questions

Every Material proposal should answer:

- Does this strengthen physical coherence?
- Does it preserve hierarchy?
- Does it improve immersion?
- Does it remain accessible?
- Would users still recognise Mosaic?
- Is this solving a behavioural problem or adding visual novelty?

If the proposal exists primarily because it "looks cool", it should be reconsidered.

Validation

Future tooling should validate:

- Material hierarchy usage
- Acrylic consistency
- Runtime Material Resolution
- Accessibility compliance
- Refraction intensity
- Platform parity

Automated validation should reinforce architectural review rather than replace it.

Governance Workflow

[mermaid]
flowchart TD

```
Proposal
Proposal --> ExistingMaterialReview
ExistingMaterialReview --> Reuse["Reuse?"]
Reuse["Reuse?"] -->|Yes| ExistingMaterial
Reuse["Reuse?"] -->|No| ArchitectureReview
ArchitectureReview
ArchitectureReview --> AccessibilityReview
AccessibilityReview --> RuntimeReview
RuntimeReview --> Approval
Approval --> Implementation
```

Reuse should remain the preferred outcome.

The Material vocabulary should remain intentionally small.

Success Criteria

The Material System succeeds when:

- users immediately recognise Mosaic,
- Acrylic feels premium rather than decorative,
- artwork appears to illuminate the interface naturally,
- runtime adaptation remains subtle,
- every client shares one physical language,
- contributors naturally reuse existing materials.

The strongest Material System is one users feel rather than consciously notice.

Architectural Decisions

ADR	Decision
ADR-106	Materials are treated as physical behaviours rather than visual effects.
ADR-107	Acrylic is the primary physical material of Mosaic.
ADR-108	Runtime Atmosphere influences Materials rather than components directly.
ADR-109	Accessibility always has higher authority than material fidelity.
ADR-110	Extensions inherit the Material System rather than extending it.

Review Status

Status

Draft

Next File

12-adr s .md

Architectural Decision Records

Purpose

The Architectural Decision Records (ADRs) contained within MDS-003 preserve the reasoning behind the Mosaic Material System.

Where previous specifications established:

- Design Tokens
- Colour
- Runtime Atmosphere

this specification establishes the physical language through which those systems become a believable environment.

These ADRs explain why Mosaic deliberately models materials as physical behaviours rather than decorative effects.

Future contributors should consult these records before proposing changes to the Material System.

ADR Format

Every Mosaic ADR follows the standard structure.

[text]

ADR Number

Status

Context

Decision

Consequences

Alternatives Considered

Related Specifications

Each ADR documents one significant architectural decision.

ADR-111

Title

Treat Materials As Physical Behaviours

Status

Accepted

Context

Most UI frameworks model materials as visual styles.

Founder workshops consistently described the desired experience as physical rather than decorative.

Decision

Materials become behavioural objects.

Rendering becomes their implementation.

Consequences

Future rendering technologies may evolve without changing the perceived physical language of Mosaic.

ADR-112

Title

Adopt Acrylic As The Primary Interactive Material

Status

Accepted

Context

Glass interfaces frequently disappear into the background while opaque panels isolate themselves from surrounding content.

Neither behaviour aligned with the desired Mosaic experience.

Decision

Premium Acrylic becomes the primary interactive material.

Consequences

The interface gains:

- presence
- depth
- environmental lighting
- restrained translucency

without competing with entertainment.

ADR-113

Title

Treat Artwork As Environmental Light

Status

Accepted

Context

Direct artwork recolouring weakens hierarchy and brand identity.

Decision

Artwork becomes a conceptual light source rather than a colour palette.

Consequences

Runtime Atmosphere influences Materials rather than directly colouring components.

This creates significantly stronger physical coherence.

ADR-114

Title

Introduce UV-Indexed Refraction

Status

Accepted

Context

Screen-space atmospheric effects become inconsistent across devices and layouts.

Decision

Atmospheric lighting is projected into a normalised UV coordinate space sampled by materials.

Consequences

Atmosphere becomes:

- device independent
- composition aware
- runtime efficient

while preserving one coherent lighting model.

ADR-115

Title

Separate Refraction From Colour

Status

Accepted

Context

Many rendering systems treat refraction as colour overlays or decorative bloom.

Decision

Refraction becomes light transport rather than colour replacement.

Consequences

Materials appear physically illuminated rather than digitally tinted.

ADR-116

Title

Material Resolution Owns Rendering Decisions

Status

Accepted

Context

Allowing components to construct their own material behaviour fragments the Material System.

Decision

Components request Material Identity.

The Runtime Material Resolver determines implementation.

Consequences

Applications remain simple while rendering technology evolves independently.

ADR-117

Title

Accessibility Overrides Material Fidelity

Status

Accepted

Context

Highly expressive material systems frequently reduce readability.

Decision

Accessibility possesses higher authority than physical realism.

Consequences

Material richness automatically adapts whenever readability would otherwise decrease.

ADR-118

Title

Maintain One Shared Environmental Lighting Model

Status

Accepted

Context

Per-component lighting creates inconsistent physical behaviour.

Decision

Every material samples one shared Runtime Atmosphere field.

Consequences

The interface feels like one coherent physical environment rather than unrelated visual effects.

ADR-119

Title

Extensions Inherit The Material System

Status

Accepted

Context

Allowing plugins to implement independent materials fragments product identity.

Decision

Extensions contribute:

- artwork
- information
- relationships

The platform owns every material behaviour.

Consequences

Community extensions automatically inherit future material improvements.

ADR Relationships

[mermaid]
flowchart TD

ADR111["Physical Behaviour"]

ADR112["Acrylic"]

ADR113["Artwork Light"]

ADR114["UV Refraction"]

ADR115["Light Transport"]

ADR116["Material Resolver"]

ADR117["Accessibility"]

ADR118["Shared Environment"]

ADR119["Extensions"]

ADR111 --> ADR112

ADR112 --> ADR115

ADR113 --> ADR114

ADR114 --> ADR115

ADR115 --> ADR118

ADR118 --> ADR116

ADR117 --> ADR116

ADR116 --> ADR119

Together these decisions establish one coherent physical model for the entire Mosaic interface.

Future ADRs

Future Material ADRs are expected to formalise:

- Spectral Refraction
- HDR Material Response
- Dynamic Edge Illumination
- Adaptive Thickness
- Volumetric Acrylic
- Ray-Marched Atmosphere
- Material Level-of-Detail

- Hardware Capability Profiles

These intentionally remain outside the scope of MDS-003 Version 0.1.

ADR Governance

Material ADRs should change only when:

- architectural inconsistencies emerge,
- runtime evolution requires refinement,
- accessibility research identifies deficiencies,
- the Design Language itself evolves.

Implementation techniques should never drive architectural changes.

Rendering engines evolve.

Material behaviour should remain recognisably Mosaic.

Summary

The ADRs contained within MDS-003 define the physical identity of Mosaic.

Rather than treating materials as visual styling, Mosaic treats them as environmental behaviours.

Light moves.

Materials respond.

Entertainment quietly illuminates the interface.

Together these decisions create a material language that feels:

- premium,
- restrained,
- believable,
- unmistakably Mosaic.

Review Status

Status

Draft

Next File

13-contributor-guidance.md

Contributor Guidance

Purpose

The Material System is one of the most distinctive parts of Mosaic.

It defines how the interface feels rather than simply how it looks.

Every contributor working on:

- rendering,
- frontend,
- design,
- shaders,
- components,
- runtime systems,

will eventually interact with this specification.

The purpose of this chapter is to ensure that every implementation strengthens one coherent physical language rather than introducing isolated visual effects.

Think In Materials

Never begin with:

How should this shader work?

Instead ask:

What physical behaviour am I trying to communicate?

Rendering should always emerge from material behaviour.

Not the reverse.

Think In Light

One useful mental model is to stop thinking about colour entirely.

Instead imagine:

Light

↓

Material

↓

User

Ask:

- Where is the light coming from?
- How should this material respond?
- Should the user notice the light...

or simply feel it?

This mindset naturally produces calmer interfaces.

Materials Before Effects

Avoid implementing:

- blur
- glow
- bloom
- gradients
- transparency

as independent effects.

Instead ask:

Which material should produce this behaviour?

Every visual effect should belong to a Material.

Not to a component.

Preserve Material Identity

Every Material possesses one responsibility.

Canvas.

Environment.

Surface.

Grouping.

Acrylic.

Physical presence.

Hero.

Current importance.

Overlay.

Temporary interaction.

Avoid blending these responsibilities together.

The Material vocabulary should remain intentionally small.

Let Atmosphere Travel

Applications should never manually tint materials.

Preferred.

[text]
Artwork



Runtime Atmosphere



Material



Presentation

Avoid.

[text]
Artwork



Component Colour

Atmosphere belongs to the Material System.

Components should remain unaware of its implementation.

Respect The Hero

The Hero Material should receive the richest physical behaviour.

However...

It should never become more visually interesting than the entertainment itself.

When implementing Hero Materials ask:

Does this strengthen the artwork...

or compete with it?

If the answer is competition...

Reduce the effect.

Protect The Canvas

The Canvas should remain calm.

Avoid introducing:

- strong gradients

- heavy atmosphere
- animated backgrounds
- decorative textures

The Canvas exists to support.

Never to perform.

Acrylic Should Feel Solid

When implementing Acrylic, imagine a physical object.

Not a CSS filter.

Acrylic should communicate:

- thickness
- diffusion
- restrained translucency
- subtle edge lighting

If the material begins feeling like glass...

Reduce transparency.

Increase perceived substance.

Refraction Exists For Understanding

Refraction should always explain physical presence.

It should never exist because shaders are available.

Ask:

Does this strengthen perceived depth?

If not...

Remove it.

The best refraction is often the one users never consciously notice.

Runtime Owns Materials

Applications should never construct Material behaviour.

Components request:

[text]
`Material.Hero`

The Runtime Material Resolver determines:

- atmosphere
- refraction
- diffusion
- edge behaviour
- lighting

This separation dramatically simplifies application code.

Accessibility Is Architectural

Accessibility is not an optional rendering mode.

Every Material implementation should begin by asking:

Will this remain readable?

Only afterwards ask:

Will this feel immersive?

Physical realism always remains subordinate to understanding.

Plugins

Extensions should never define:

- Acrylic
- Blur
- Refraction
- Lighting
- Materials

Instead they contribute:

- artwork
- information
- relationships

The Material System automatically integrates those contributions into one coherent physical environment.

Device Independence

Do not optimise Materials for one platform.

Instead ask:

Would this still feel like Acrylic on:

- Desktop?
- Mobile?
- Television?
- Future devices?

Implementation may differ.

Perception should remain constant.

Performance

Performance optimisation should preserve behaviour.

Preferred.

[text]
Same Material Behaviour

↓

Simpler Implementation

Avoid.

[text]
Different Behaviour

↓

Higher Performance

If simplification becomes necessary:

Reduce implementation complexity.

Preserve conceptual behaviour.

Common Mistakes

Avoid the following.

Frosted Glass

Blur replacing material behaviour.

Decorative Glow

Glow existing independently from Runtime Atmosphere.

Transparent Panels

Interfaces becoming visually fragile.

Material Per Component

Every component inventing independent physical behaviour.

Heavy Atmosphere

Materials becoming more expressive than entertainment.

Platform-Specific Behaviour

Desktop Acrylic behaving fundamentally differently from Television Acrylic.

Material Review Questions

Before implementing any Material ask:

- What physical object does this represent?
- Where does its light originate?
- Does this preserve hierarchy?
- Does this strengthen immersion?
- Would removing this effect reduce understanding?
- Does this still feel like Mosaic?

If the final answer is uncertain...

Return to the Material hierarchy before writing code.

Material Checklist

Every Material implementation should satisfy the following.

- ☐ Material identity is clear.
- ☐ Runtime Atmosphere is respected.
- ☐ Refraction remains subtle.
- ☐ Accessibility remains uncompromised.
- ☐ Components remain unaware of implementation.
- ☐ Device independence is preserved.
- ☐ Performance optimisations preserve behaviour.
- ☐ Entertainment remains visually dominant.

Final Guidance

The Material System exists to make software feel less like software.

When contributors find themselves discussing:

- shaders,
- blur,
- transparency,
- rendering,

they should pause and instead ask:

What should this material feel like?

If that question becomes habitual, the implementation will naturally begin aligning with the Mosaic Design Language.

The strongest Material System is not the most technically impressive.

It is the one users stop noticing because it feels completely natural.

Review Status

Status

Draft

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glossary.md

Glossary

Purpose

This glossary defines the architectural terminology introduced by **MDS-003 — Material System**.

Unlike previous glossaries, this document focuses on the physical behaviour of the Mosaic interface rather than conceptual behaviour or colour.

These definitions should remain stable regardless of rendering technology.

Future specifications should reuse these terms consistently.

A

Acrylic

The primary interactive material within the Mosaic Material System.

Acrylic is a semi-translucent material that receives:

- Runtime Atmosphere
- Refraction
- Environmental Light

while preserving:

- hierarchy
- readability
- physical presence

Acrylic is intentionally distinct from glass.

It possesses substance.

Atmosphere

The environmental lighting generated from the user's current entertainment.

Atmosphere is received by Materials.

It is not applied directly to components.

C

Canvas

The foundational environmental material beneath every Mosaic Composition.

Canvas communicates:

- calmness
- stability
- continuity

It receives minimal atmospheric influence.

D

Diffusion

The soft internal spreading of environmental light through a material.

Diffusion reduces:

- saturation
- contrast
- visual noise

while increasing perceived physical realism.

E

Edge Lighting

The subtle illumination visible along the edges of Acrylic materials.

Edge Lighting communicates:

- thickness
- craftsmanship
- physical presence

It should remain understated.

H

Hero Material

The material representing the highest-priority concept within the current Composition.

Hero Material receives the strongest Runtime Atmosphere and Refraction while preserving the dominance of entertainment artwork.

L

Light Transport

The conceptual movement of environmental light through the Material System.

Light Transport describes behaviour rather than implementation.

It should feel physically believable rather than visually dramatic.

M

Material

A physical behavioural object within the Mosaic Design System.

Materials communicate:

- hierarchy
- environmental response
- physical presence

They are not equivalent to rendering effects.

Material Hierarchy

The ordered relationship between all Mosaic materials.

[text]
Canvas

↓

Surface

↓

Acrylic

↓

Hero

↓

Overlay

Each level communicates increasing physical presence.

Material Resolution

The runtime process that transforms Material Identity into concrete rendering behaviour.

Material Resolution evaluates:

- Runtime Atmosphere

- Accessibility
- Device Capability
- Theme

before producing renderable material properties.

O

Overlay Material

A temporary material prioritising interaction and readability.

Overlay Material participates in the Material System while reducing atmospheric influence.

It exists only while interaction requires it.

P

Physical Presence

The perceived solidity of a material.

Physical Presence is communicated through:

- diffusion
- edge lighting
- depth
- refraction

rather than shadows or opacity alone.

R

Refraction

The controlled transport and bending of environmental light through Mosaic materials.

Refraction communicates:

- depth
- atmosphere
- environmental continuity

It should never become decorative.

Runtime Material Resolver

The platform subsystem responsible for producing resolved Material Profiles.

Applications consume resolved materials rather than constructing them.

S

Surface

A grouping material sitting between Canvas and Acrylic.

Surfaces organise information while remaining visually restrained.

They receive limited atmospheric influence.

T

Thickness

The perceived depth of a material.

Thickness is communicated through:

- edge behaviour
- diffusion
- refraction
- lighting

rather than explicit dimensional rendering.

U

UV Field

The normalised environmental lighting space generated from the current Hero.

Materials sample this shared field to produce coherent atmospheric behaviour.

The UV Field is independent of:

- screen resolution
- layout
- rendering technology

UV-Indexed Refraction

The conceptual lighting model through which Runtime Atmosphere is projected into normalised UV space before being sampled by Materials.

This system provides one coherent environmental lighting model across every client.

Cross References

Specification	Primary Concepts
MDL-001 Vision	Companion, Immersion
MDL-002 Principles	Calmness, Restraint
MDL-003 Mental Model	World, Focus
MDL-004 Interaction Model	Continuity
MDL-005 Composition Model	Hero, Hierarchy
MDS-001 Design Token Architecture	Runtime Tokens
MDS-002 Colour System	Runtime Atmosphere

Terminology Rules

Future contributors should:

- describe materials as behaviours rather than effects
- describe light before colour
- distinguish Acrylic from glass
- distinguish Atmosphere from Brand
- distinguish Refraction from blur

Material terminology should remain independent from rendering implementation.

Review Status

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references.md

References

Purpose

This document records the architectural influences and conceptual foundations that informed **MDS-003 — Material System**.

Unlike implementation documentation, these references explain *why* the Material System behaves as it does rather than prescribing rendering techniques.

The Material System intentionally combines ideas from:

- physical materials,
- environmental lighting,
- industrial design,
- human perception,
- rendering architecture,
- entertainment interfaces,

into one coherent architectural language.

Reading Order

Contributors should approach references in the following order.

1. MDL Specifications 2. Design Token Architecture 3. Colour System 4. Material System 5. Rendering Architecture 6. Platform Implementations

The Mosaic Design Language remains the authoritative source.

External references provide inspiration rather than specification.

Internal References

MDL-001 — Vision

Provides:

- Companion philosophy
- Entertainment-first thinking
- Premium product identity

The Material System exists to reinforce immersion without competing with entertainment.

MDL-002 — Principles

Provides:

- Content Leads
- Every Feature Earns Its Place
- Calm Interfaces
- Context Before Prediction

Materials should quietly support these principles rather than becoming visual features in their own right.

MDL-003 — Mental Model

Provides:

- World
- Focus
- Context
- Information
- Relationships

Materials physically express the World.

They never redefine it.

MDL-004 — Interaction Model

Provides:

- Continuity
- Behaviour
- Movement
- Temporal evolution

Material behaviour should evolve alongside interaction rather than independently from it.

MDL-005 — Composition Model

Provides:

- Hero
- Hierarchy
- Priority
- Density
- Anchors

Material Hierarchy is intentionally derived from the Composition Model.

Physical presence should reinforce conceptual hierarchy.

MDS-001 — Design Token Architecture

Provides:

- Semantic Tokens
- Runtime Tokens
- Resolution Pipeline

Materials consume Runtime Tokens rather than generating their own semantic meaning.

MDS-002 — Colour System

Provides:

- Brand
- Semantic Colours
- Runtime Atmosphere
- Theme Architecture

The Material System intentionally treats Runtime Atmosphere as environmental light rather than surface colour.

Future Specifications

The following specifications directly depend upon MDS-003.

- MDS-004 Typography
- MDS-005 Motion System
- MDS-006 Composition Engine
- MDS-007 Tile Framework
- MDS-008 Component Library

These specifications should consume Materials rather than introducing independent physical behaviour.

Industrial Design

The Material System draws significant inspiration from physical industrial materials rather than digital interface trends.

Examples include:

- polished acrylic
- machined aluminium
- architectural glass
- illuminated resin

- museum display systems

The objective is not literal reproduction.

It is perceived physical credibility.

Environmental Lighting

One of the defining conceptual influences is environmental lighting.

Instead of colouring interface objects directly:

Light

↓

Materials

↓

Perception

This model creates stronger visual coherence than direct colour application.

It is one of the defining architectural ideas behind Mosaic.

Human Perception

Several characteristics of visual perception influenced the Material System.

Examples include:

- depth perception
- edge recognition
- environmental lighting
- peripheral vision
- object permanence

The Material hierarchy is designed to reinforce these perceptual behaviours rather than work against them.

Rendering Architecture

Although MDS intentionally avoids implementation details, the Material System assumes future rendering architectures capable of:

- GPU acceleration
- material composition
- runtime atmosphere
- cached lighting
- adaptive shaders

These technologies remain implementation concerns.

The Material behaviour defined within MDS should remain stable regardless of rendering backend.

Physical Simulation

The Material System is **not** intended to become a physically accurate rendering engine.

Instead it follows a principle of:

Perceived realism over physical realism.

Every rendering decision should improve:

- understanding
- immersion
- calmness

rather than simulation accuracy.

Runtime Behaviour

Materials intentionally evolve with:

- Focus
- Context
- Runtime Atmosphere
- Theme
- Accessibility

The Material System therefore treats rendering as a continuously evolving environmental system rather than a static skin.

Platform Independence

Material behaviour should remain consistent across:

- Web
- Flutter
- SwiftUI
- Jetpack Compose
- Future rendering engines

Implementation quality may differ.

Material identity should not.

Mosaic-Specific Influences

The Material System emerged directly from founder exploration.

Major architectural discoveries included:

- Acrylic should feel substantial rather than transparent.
- Artwork should illuminate the interface rather than recolour it.
- Refraction should communicate physical presence rather than visual effects.
- Runtime Atmosphere should behave like environmental lighting.
- Materials should support entertainment rather than compete with it.
- UV-indexed lighting provides stronger conceptual consistency than screen-space effects.

Together these discoveries define the physical language unique to Mosaic.

Relationship To The Refraction System

The Material System intentionally positions Refraction as a consequence of:

[text]
Artwork

↓

Runtime Atmosphere

↓

Material

↓

Refraction

↓

Presentation

This ordering ensures:

- hierarchy remains stable,
- brand identity remains recognisable,
- materials remain physically believable.

Future rendering implementations should preserve this architectural ordering.

Normative References

Required reading before contributing to MDS-003.

- MDL-001 Vision
- MDL-002 Principles
- MDL-003 Mental Model

- MDL-004 Interaction Model
- MDL-005 Composition Model
- MDS-001 Design Token Architecture
- MDS-002 Colour System

Together these documents define the conceptual foundation of the Material System.

Informative References

Future contributors may also wish to review:

- MDS-005 Motion System
- MDS-006 Composition Engine
- MDS-007 Tile Framework
- MDS-008 Component Library

These specifications describe how Material behaviour becomes interaction and presentation.

Living Document

This reference list should remain intentionally concise.

References should only be introduced when they materially influence:

- material behaviour,
- environmental lighting,
- runtime architecture,
- implementation boundaries.

The objective is to preserve architectural reasoning rather than catalogue rendering techniques.

Completion

This concludes **MDS-003 — Material System**.

The next specification in the Mosaic Design System is:

MDS-004 — Typography System

Where MDS-003 defines **how the interface physically exists**, MDS-004 defines **how language exists within that physical world**.

It formalises:

- Typography philosophy
- Reading hierarchy
- Editorial rhythm

- Responsive type scales
- Reading density
- Hero typography
- Motion-aware typography
- Accessibility
- Cross-platform typography

Together with the Material System, Typography completes the primary sensory language of Mosaic.